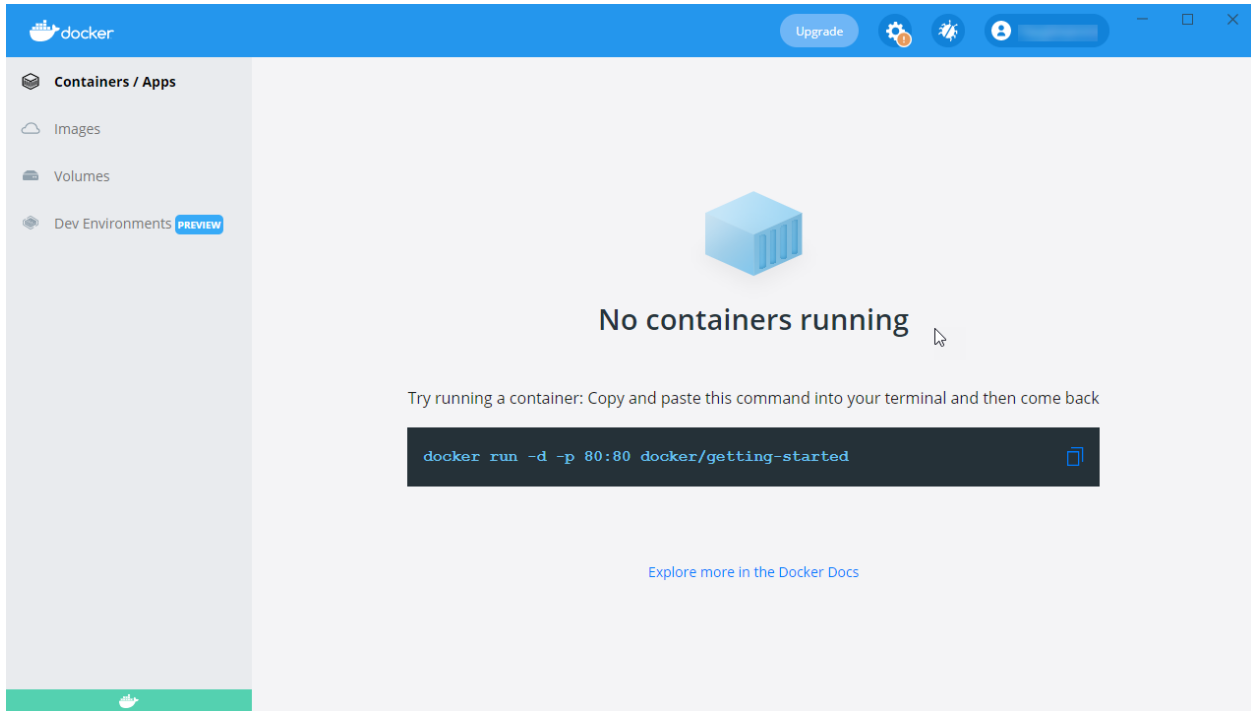


Final Assignment 14.1: Java and Debezium

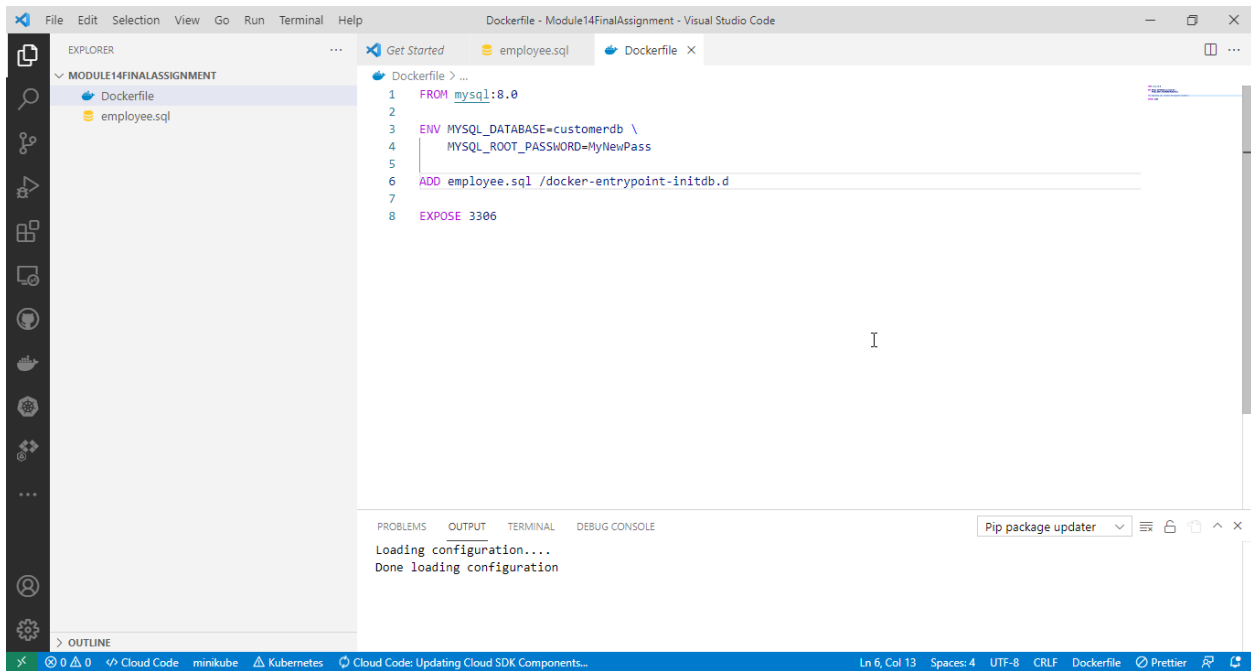
1.



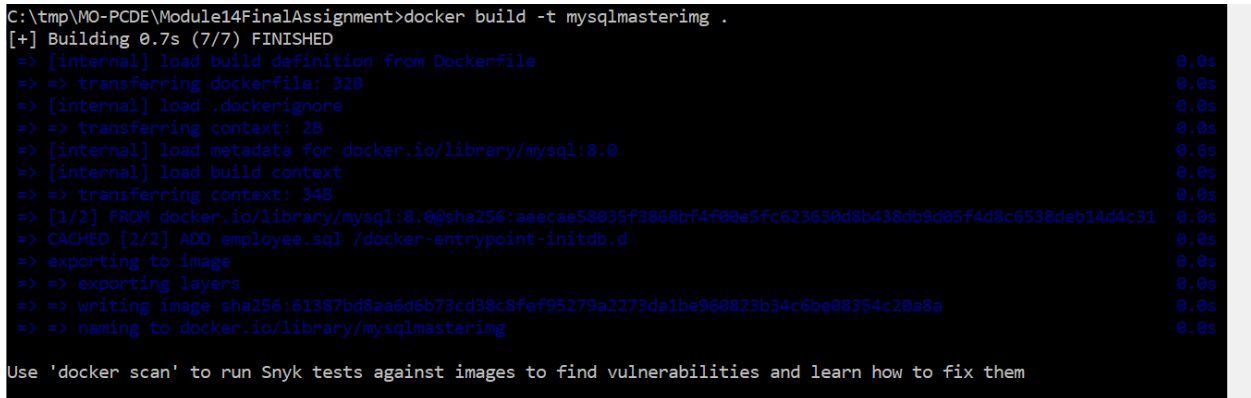
2.

```
C:\tmp\MO-PCDE>docker network create CDCFinalAssignment
59d54f1cd08b2b2a9115997428ca9fe3052bef76403dfa39fd92052eeba2daba
```

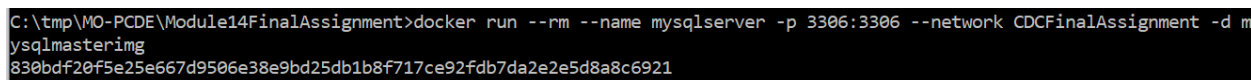
3.



4.



5.



6.

 mysqlserver [mysqlmastering](#)
RUNNING PORT: 3306

7.

```
C:\tmp\MO-PCDE\Module14FinalAssignment\debezium-main>dir
Volume in drive C is OS
Volume Serial Number is 92CD-86B9

Directory of C:\tmp\MO-PCDE\Module14FinalAssignment\debezium-main

11/21/2021  11:30 AM  <DIR>          .
11/21/2021  11:30 AM  <DIR>          ..
06/22/2021  02:27 PM  <DIR>          app
06/22/2021  02:27 PM                74 Dockerfile
06/22/2021  02:27 PM            1,063 LICENSE
06/22/2021  02:27 PM                45 README.md
               3 File(s)            1,182 bytes
               3 Dir(s)  58,764,144,640 bytes free
```

8.

```
C:\tmp\MO-PCDE\Module14FinalAssignment\debezium-main>docker build -t debeziumimg .
[+] Building 0.4s (8/8) FINISHED
=> [internal] load build definition from Dockerfile                                0.0s
=> => transferring dockerfile: 111B                                              0.0s
=> [internal] load .dockerignore                                                  0.0s
=> => transferring context: 2B                                                    0.0s
=> [internal] load metadata for docker.io/library/maven:3.6.3-openjdk-11        0.0s
=> [internal] load build context                                                  0.1s
=> => transferring context: 24.96kB                                              0.0s
=> [1/3] FROM docker.io/library/maven:3.6.3-openjdk-11                        0.2s
=> [2/3] COPY app /tmp/                                                          0.0s
=> [3/3] WORKDIR /tmp/                                                          0.0s
=> exporting to image                                                            0.1s
=> => exporting layers                                                            0.0s
=> => writing image sha256:ba77d315881654c36e459d2e00663af05a288c47dc4ea86b886a1416bef648fd 0.0s
=> => naming to docker.io/library/debeziumimg                                  0.0s

Use 'docker scan' to run Snyk tests against images to find vulnerabilities and learn how to fix them
```

9.

 debeziumserver [debeziumimg](#)
RUNNING

10.

```
docker exec -it 5bd420feefa6c1a9673bd0105cfd5b4a44e6eef31733838d1290f100e13bfbf1 /bin/sh
# apt-get update
Get:1 http://deb.debian.org/debian buster InRelease [122 kB]
Get:2 http://security.debian.org/debian-security buster/updates InRelease [65.4 kB]
Get:3 http://deb.debian.org/debian buster-updates InRelease [51.9 kB]
Get:4 http://security.debian.org/debian-security buster/updates/main amd64 Packages [309 kB]
Get:5 http://deb.debian.org/debian buster/main amd64 Packages [7906 kB]
Get:6 http://deb.debian.org/debian buster-updates/main amd64 Packages [15.2 kB]
Fetched 8469 kB in 2s (5092 kB/s)
Reading package lists... Done
# apt-get install nano
Reading package lists... Done
Building dependency tree
Reading state information... Done
Suggested packages:
  spell
The following NEW packages will be installed:
  nano
0 upgraded, 1 newly installed, 0 to remove and 21 not upgraded.
Need to get 544 kB of archives.
After this operation, 2269 kB of additional disk space will be used.
Get:1 http://deb.debian.org/debian buster/main amd64 nano amd64 3.2-3 [544 kB]
Fetched 544 kB in 0s (5702 kB/s)
debconf: delaying package configuration, since apt-utils is not installed
Selecting previously unselected package nano.
(Reading database ... 12572 files and directories currently installed.)
Preparing to unpack .../archives/nano_3.2-3_amd64.deb ...
Unpacking nano (3.2-3) ...
Setting up nano (3.2-3) ...
update-alternatives: using /bin/nano to provide /usr/bin/editor (editor) in auto mode
update-alternatives: using /bin/nano to provide /usr/bin/pico (pico) in auto mode
#
```

11.

```
Command Prompt - docker run -it --rm --name debeziumserver --network CDCFinalAssignment debeziumimg bash
GNU nano 3.2 /tmp/src/main/java/mit/edu/tv/config/DebeziumConnectorConfig.java

return io.debezium.config.Configuration.create()
    .with("name", "customer-mysql-connector")
    .with("connector.class", "io.debezium.connector.mysql.MySqlConnector")
    .with("offset.storage", "org.apache.kafka.connect.storage.FileOffsetBackingStore")
    .with("offset.storage.file.filename", offsetFile)
    .with("offset.flush.interval.ms", "60000")
    .with("database.hostname", "mysqlserver")
    .with("database.port", 3306)
    .with("database.user", "root")
    .with("database.password", "MyNewPass")
    .with("database.dbname", "employeeeb")
    .with("database.include.list", "employeeeb")
    .with("include.schema.changes", "false")
    .with("database.allowPublicKeyRetrieval", "true")
    .with("database.server.id", "10181")
    .with("database.server.name", "localhost_employeeeb")
    .with("database.history", "io.debezium.relational.history.FileDatabaseHistory")
    .with("database.history.file.filename", historyFile)
    .build();
}

^G Get Help      ^O Write Out    ^W Where Is     ^K Cut Text     ^J Justify      ^C Cur Pos     M-U Undo       M-A Mark Text
^X Exit          ^R Read File    ^_ Replace      ^U Uncut Text   ^I To Spell     ^_ Go To Line   M-E Redo       M-G Copy Text
```

12.

```
2021-11-21 17:48:07.464 INFO 469 --- [ysqlserver:3306] io.debezium.util.Threads : Creating thread debezium-mysqlconnector-localhost_employeeedb-binlog-client
Key = 'Struct{id=1}' value = 'Struct{after=Struct{id=1,FirstName=John,LastName=Doe,MobileNumber=1112223333,email=john@doe.com},source=Struct{version=1.5.2.Final,connector=mysql,name=localhost_employeeedb,ts_ms=1637516887429,snapshot=last,db=employeeedb,table=employee,server_id=0,file=binlog.000002,pos=156,row=0},op=r,ts_ms=1637516887431}'
2021-11-21 17:48:07.563 INFO 469 --- [rce-coordinator] i.d.c.m.MySqlStreamingChangeEventSource : Keepalive thread is running
```

13.

```
# mysql -h localhost -u root -pMyNewPass employeeedb
mysql: [Warning] Using a password on the command line interface can be insecure.
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 8
Server version: 8.0.27 MySQL Community Server - GPL

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> show tables;
```

14.

```
mysql> INSERT INTO employeeedb.employee VALUES (2, "Mary", "Doe", "4351234354", "mary@doe.com");
Query OK, 1 row affected (0.02 sec)

mysql>
```

15.

```
2021-11-21 17:56:01.023 INFO 469 --- [pool-1-thread-1] i.d.connector.common.BaseSourceTask : 2 records sent during previous 00:07:54.387, last recorded offset: {transaction_id=null, ts_sec=1637517360, file=binlog.000002, pos=235, row=1, server_id=1, event=2}
Key = 'Struct{id=2}' value = 'Struct{after=Struct{id=2,FirstName=Mary,LastName=Doe,MobileNumber=4351234354,email=mary@doe.com},source=Struct{version=1.5.2.Final,connector=mysql,name=localhost_employeeedb,ts_ms=1637517360000,db=employeeedb,table=employee,server_id=1,file=binlog.000002,pos=393,row=0},op=c,ts_ms=1637517360824}'
```